Geophysics III

Complex Numbers Tutorial

"You would have to be half mad to dream me up."
— Lewis Carroll, *Alice in Wonderland*

1 Number Sets

Before we go down the rabbit hole, I need to introduce you to a few bits of terminology. Mathematicians especially, but physicists also use these terms and notation as shorthand to identify groups of numbers with certain characteristics. These are identified in Table 1. The largest number set is the set of all complex numbers which includes all combinations of real and imaginary numbers (Figure 1).

Table 1: Number Sets

Notation	Number set
$\mathbb{N}$	Natural (counting) numbers: $1, 2, 3, \ldots$
$\mathbb{Z}$	Integers: $\ldots -3, -2, -1, 01, 2, 3, \ldots$; includes $\mathbb{N}$
$\mathbb{R}$	Real numbers: all rational and irrational numbers, $(-\infty, \infty)$; rational numbers include all integers and numbers that can be expressed as one integer divided by another, while irrational numbers are the numbers which cannot be expresses as integer ratios (e.g., π, e, $\sqrt{2}$); includes $\mathbb{N}$ and $\mathbb{Z}$
$\mathbb{R}^+$	Real positive numbers, $(0, \infty)$
$\mathbb{R}_0^+$	Real positive numbers and zero, $[0, \infty)$
$\mathbb{I}$	Imaginary numbers: all numbers that can be expressed as a real number times $\sqrt{-1}$, i.e. numbers that when squared give a negative, rather than positive, number
$\mathbb{C}$	Complex numbers: All numbers containing a real ($\mathbb{R}$) and an imaginary part ($\mathbb{I}$)

2 Introduction

Mathematicians have problems with things that are undefined. I can't blame them. There is something not satisfying when trying to compute a number and instead you run up against something that is not allowed under the current set of rules. So what is a mathematician do?

Make new rules and see where they lead. Take for example the square root function. In a more classical scheme, the square root function is typically defined for positive numbers, i.e.,

$$y = \begin{cases} \sqrt{x} & x \geq 0 \\ \text{undefined} & \text{x} < 0 \end{cases},$$

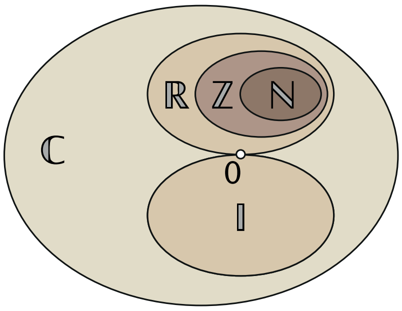

Figure 1: Relationship between the number sets listed in Table 1. Sets such as $\mathbb{N}$ are entirely captured by $\mathbb{Z}$. All numbers, both real, $\mathbb{R}$ and purely imaginary, $\mathbb{I}$, as well as any arbitrary combination of the two, are included in the set of complex numbers, $\mathbb{C}$. The $\mathbb{R}$ and $\mathbb{I}$ sets intersect only at 0 (white point) and therefore are orthogonal.

where $x, y \in \mathbb{R}$. But what happens when x is negative? For example, imagine the case when $y = \sqrt{-1}$. Mathematicians long ago considered this question, but initially found no significance beyond mere curiosity; hence, the name imaginary was given to this group of numbers. But as it turns out, they can be used as a compact way to describe waves (we're a bit far from that at the moment, but we'll get there).

3 Notation

We denote the imaginary number by i, defined as

$$i \equiv \sqrt{-1}. \tag{1}$$

Hence, $i^2 = -1$. We can write any negative root as

$$\begin{aligned}\sqrt{-b^2} &= \sqrt{(-1)(b^2)},\\ &= b\sqrt{-1},\\ &= bi,\end{aligned}$$

where $b \in \mathbb{R}$. What this tells us is that we can represent an imaginary number as *the* imaginary number, i, multiplied by a real number.

We can now extend our definition of the square root to include the root of negative numbers,

$$y = \begin{cases}\sqrt{x} & x \geq 0\\ i\sqrt{|x|} & x < 0\end{cases},$$

where $x \in \mathbb{R}$, $|x|$ is the absolute value of x, and $y \in \mathbb{C}$.

Although this tutorial covers complex numbers, dealing with them is not necessarily complex. In fact, if you are familiar and comfortable with vectors, dealing with complex numbers is in some ways very similar.

A complex number has both a real part and a purely imaginary part and can be written in the form,

$$\overbrace{z}^{\text{complex number}} = \underbrace{a}_{\text{real part}} + \underbrace{bi}_{\text{imaginary part}} \tag{2}$$

where $a, b \in \mathbb{R}$. Just like vector components, the real part and imaginary part cannot be combined. There are several other ways to represent a complex number and we'll get to those in a bit.

4 The Complex Plane

Imagine an infinite line (no artists here, it must be straight). On this line, we'll add a zero. It doesn't matter where we place it since the line is infinite on either side, it just gives us a point of reference from which to define our positive and negative directions. If we make everything to the right positive and everything to the left negative, we can mark out divisions at integer steps from zero. This is our real number line and represents all real numbers. Between the integers, we have the infinite number of points between including the rational and irrational numbers.

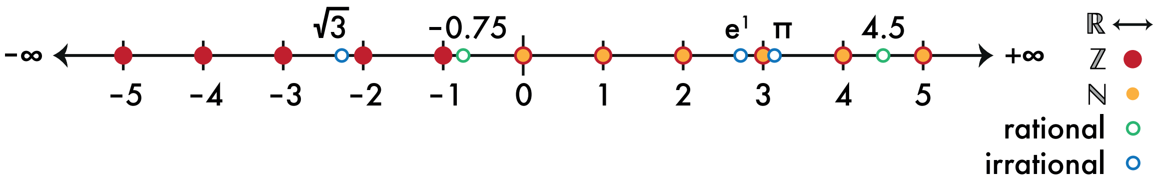

Figure 2: The real number line. Selected numbers from some of the number sets are identified. While example rational and irrational numbers are singled out of the real number line, in reality, there are an infinitely many numbers between each integer value.

Now imagine a second number line much the same, but each tick mark on this new line represents an integral step in either the positive, $+i$, or negative, $-i$, direction. In between we have all the rational and irrational imaginary numbers all the same. We have a $0i$, but 0 multiplied by any number, i included, is 0. Hence, 0 occurs on both the real and imaginary number lines indicating that they intersect. Because real and imaginary numbers cannot be added, the two number lines must be orthogonal (at right angles to each other).

So now plotting complex numbers is much like plotting (x,y) Cartesian pairs that you are already familiar with (Figure 4).

5 Math Operations involving Complex Numbers

5.1 Complex Operators

Note for the imaginary function it returns a real number b. The final operator that we will define is called the *complex conjugate* $(*)$. If $z = x + iy$, then the complex conjugate of z is

$$z^* = x - iy, \tag{3}$$

which simply changes the sign of the imaginary part. This may seem like a very silly operator, but, as you will soon see, is vital to computing the magnitude (norm) of a complex number.

At times, we want to isolate the real part. To signify that we are only using the real part we define a function, Re, which returns the real part. For example, for a complex number, z, written in the form $x + iy$, we want the function to yield

$$\mathrm{Re}(z) = x.$$

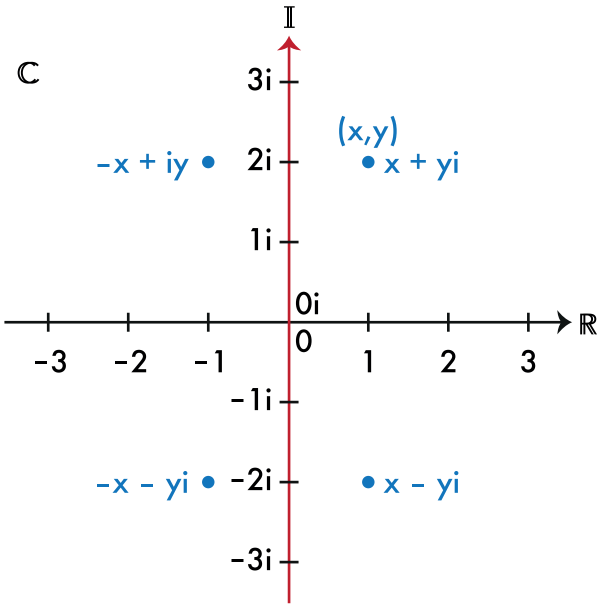

Figure 3: The complex plane, $\mathbb{C}$ includes both the real, $\mathbb{R}$, and imaginary number lines, $\mathbb{I}$, which intersect at $0 = 0i$. An example point is shown at $(x, y) = (1, 2)$.

Likewise, there we need a function that returns the imaginary part,

$$\mathrm{Im}(z) = y.$$

Each of these functions are defined in terms of the complex conjugate,

$$\mathrm{Re}(z) = \frac{z + z^*}{2} \tag{4}$$

$$\mathrm{Im}(z) = \frac{z - z^*}{2i}. \tag{5}$$

5.2 Basic Operations

Many arithmetic operations involving complex numbers are very similar to those for vectors where the real and imaginary parts are treated as separate components. Consider two complex numbers, $z_1 = a + ib$ and $z_2 = c + id$. To add and subtract z_1 and z_2,

$$\begin{aligned} z_1 + z_2 &= (a + ib) + (c + id) \\ &= (a + c) + i(b + d), \\ z_1 - z_2 &= (a + ib) - (c + id) \\ &= (a - c) + i(b - d). \end{aligned}$$

To multiply, distribute as you might a polynomial and collect terms,

$$\begin{aligned} z_1 z_2 &= (a + ib)(c + id) \\ &= ac + iad + ibc + i^2 bd \\ &= ac + iad + ibc - bd \\ &= (ac - bd) + i(ad + bc). \end{aligned}$$

To divide, we employ the complex conjugate to remove a complex number from the denominator, e.g.,

$$\begin{aligned}
\frac{z_1}{z_2} &= \frac{(a+ib)}{(c+id)} \\
\frac{z_1}{z_2}\frac{z_2^*}{z_2^*} &= \frac{(a+ib)}{(c+id)}\frac{(c-id)}{(c-id)} \\
\frac{z_1 z_2^*}{|z_2|} &= \frac{(ac+bd)+(bc-ad)i)}{(c^2+d^2)}.
\end{aligned}$$

Problem 5.1. Show

$$\frac{1}{1-i} = \frac{1+i}{2}.$$

5.3 Norm

How big is $4+1i$ and is it greater than $2+3i$? We often need to know just how big a complex number is, but how to tell? This concept will make a bit more sense once we define the complex number in a trigonometric representation (e.g., Figure 4). The size (magnitude) of a complex number is called the norm and is computed as a distance from the origin to the complex point. For a vector, this is simply done by using the Pythagorean theorem; i.e., if $v = x\hat{x} + y\hat{y}$,

$$\begin{aligned}
|v| &= (v \cdot v)^{1/2} \\
&= [(x\hat{x} + y\hat{y}) \cdot (x\hat{x} + y\hat{y})]^{1/2} \\
&= (x^2 + y^2)^{1/2}.
\end{aligned}$$

where $\hat{x}$ and $\hat{y}$ are unit vectors in the x and y directions, respectively.

We wish to define a similar norm for complex numbers, but if you were to say square each component, we would have the following problem: $x^2 + (iy)^2 = x^2 - y^2$. Note that this is the difference the squares, not the length we expected. For complex numbers, the norm is defined in terms of the number and its conjugate as

$$\begin{aligned}
|z| &= (zz^*)^{1/2} \\
&= [(x+iy)(x-iy)]^{1/2} \\
&= (x^2 - ixy + ixy - i^2y^2)^{1/2} \\
&= (x^2 - (-1)y^2)^{1/2} \\
&= (x^2 + y^2)^{1/2}.
\end{aligned}$$

6 Alternative Representations

6.1 Trigonometric Functions

The $x + iy$ form makes it easy to separate the real and imaginary parts and plot on the complex plane. However, as you will see when we get to functions, the standard form may not be the most straightforward physical description. To aid interpretation, we can write a complex number as

$$z = \overbrace{|z|}^{\text{magnitude}}(\cos \underbrace{\phi + i \sin \phi}_{\text{phase angle}}), \tag{6}$$

where ϕ is the phase angle.

This trigonometric representation comes from transformation of a complex number from $x + iy$ form to a polar coordinate reference frame. Just as we would for a Cartesian coordinate pair, we can determine the polar parameters in terms of the distance from the origin (magnitude) and an angle, which we define as counterclockwise from the real axis (Figure 4).

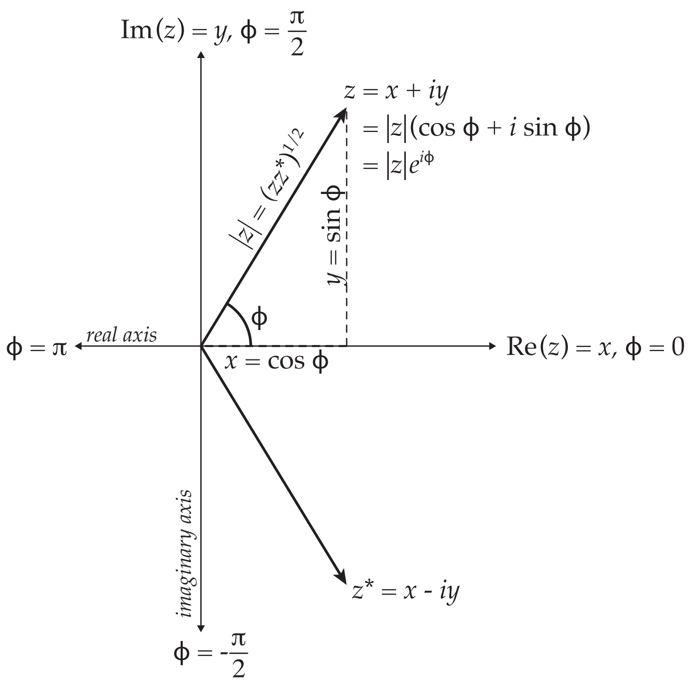

Figure 4: Complex plane and relationships between orthogonal coordinates, a circle (polar coordinates), and the natural number.

To find the magnitude, we compute the norm,

$$|z| = (x^2 + y^2)^{1/2}.$$

Note that I use the single bar, |, on either side of the variable above to denote the norm, but it is very common to see a double bar to denote the same (e.g., $\|z\|$). The $|z|$ or $\|z\|$ are used by some authors to indicate different things so it is a good idea to determine how the author defines them. It may seem as if I have defined the complex number in terms of itself—or rather it's norm—but in reality, $|z|$ is a real-valued constant or a real-valued function (i.e., a function with no complex part).

The phase is determined by the ratio of the imaginary to the real parts by

$$\phi = \tan^{-1} \frac{\mathrm{Im}\, z}{\mathrm{Re}\, z}. \tag{7}$$

By using Equations 4 and 5, we find the useful result,

$$\tan\phi = \frac{\frac{z - z^*}{2}}{\frac{z + z^*}{2}}$$
$$\tan\phi = \frac{z - z^*}{z + z^*}.$$

To convert from the trigonometric representation to $x + iy$ form,

$$x = |z|\cos\phi, \tag{8}$$
$$y = |z|\sin\phi. \tag{9}$$

Problem 6.1. Using $z = x + iy$, show that $\tan\phi = \frac{y}{x}$.

Problem 6.2. What is the complex conjugate of $z = |z|\,(\cos\phi + i\sin\phi)$?

6.2 Exponential Function

The exponential form is advantageous because it is perhaps the most compact way to write a complex number (function) and contains the same easily identifiable information content as the trigonometric form. Another reason to write a complex number in exponential form is that it tends to simplify derivatives. Recall $\frac{d}{dx}e^x = e^x$ for $x \in \mathbb{R}$. For complex exponentials, $\frac{d}{dz}e^z = e^z$ for $z \in \mathbb{C}$. More on this later.

The exponential form is given as

$$z = \overset{\text{magnitude}}{\boxed{|z|}}\, e^{i\overset{\text{phase}}{\boxed{\phi}}}, \tag{10}$$

where $|z|$ is the magnitude and ϕ is the phase angle. The complex number z need not be constant, but may be any arbitrary function. Likewise, the phase may also be a function defined as above in Equation 7.

Proof: The exponential form may seem like a long way from our $x + iy$ or the trigonometric forms so lets explore where this comes from and then we'll describe a few important values. We can represent many functions, $f(z)$, about a given point, a, by an infinite series (called a Taylor series) given by

$$f(z) = f(a) + \sum_{n=1}^{\infty} f^{(n)}(a)\frac{(z-a)^n}{n!}$$
$$f(z) = f(a) + f^{(1)}(a)(z-a) + f^{(2)}(a)\frac{(z-a)^2}{2!} + f^{(3)}(a)\frac{(z-a)^3}{3!} + \ldots + + f^{(n)}(a)\frac{(z-a)^n}{n!} + \ldots,$$

where $f^{(n)}(a)$ signifies the n-th derivative of $f(z)$ evaluated at a and $n!$ is the factorial function for n, i.e., $n! = 1\cdot 2\cdot 3\cdot\ldots n$. At $a = 0$ and when $x \in \mathbb{R}$,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \ldots.$$

Likewise, we can represent the sine an cosine functions as

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$
$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots.$$

Now suppose $x = i\phi$, the exponential function becomes

$$e^{i\phi} = 1 + i\phi + \frac{(i\phi)^2}{2!} + \frac{(i\phi)^3}{3!} + \frac{(i\phi)^4}{4!} + \frac{(i\phi)^5}{5!} + \dots.$$

If we employ the following repeating pattern from integer powers of i, i.e.,

$$\begin{aligned} i^1 &= i \\ i^2 &= -1 \\ i^3 &= -i \\ i^4 &= 1 \\ i^5 &= i \\ &\vdots, \end{aligned}$$

we find

$$e^{i\phi} = 1 + i\phi - \frac{\phi^2}{2!} - \frac{i\phi^3}{3!} + \frac{\phi^4}{4!} + \frac{i\phi^5}{5!} - \dots.$$

Using the same relationships above for sine and cosine ($x = \phi$), we find

$$\cos\phi + i\sin\phi = 1 + i\phi - \frac{x^2}{2!} - \frac{i\phi^3}{3!} + \frac{x^4}{4!} + \frac{i\phi^5}{5!} - \dots.$$

And thus it is shown that

$$e^{i\phi} = \cos\phi + i\sin\phi. \tag{11}$$

Q.E.D.

Problem 6.3. Using the trigonometric representation, show

$$\begin{aligned} e^{i0} &= 1 \\ e^{i\frac{\pi}{4}} &= \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \\ e^{i\frac{\pi}{2}} &= i \\ e^{i\pi} &= -1 \\ e^{i\frac{3\pi}{2}} &= -i \\ e^{in\pi} &= \begin{cases} 1 & n \text{ is even} \\ -1 & n \text{ is odd} \end{cases} \end{aligned}$$

Problem 6.4. What is the complex conjugate of $e^{i\phi}$?

7 Complex Functions

Some of the functions that we will examine in class are complex functions (specifically waves). A function similar to one we will discuss in class can be written

$$E_x(z) = E_0 \exp\left[-\frac{(1+i)}{\sqrt{2}}z\right]. \tag{12}$$

In this example, z is a spatial coordinate and no longer a complex number. The function is shown in Figure 5. The resultant function is a spiral in the complex plane that exponentially approaches zero as z increases. Functions in 3-dimensional plots can be relatively difficult to interpret so we often represent them parametrically[1]. In the case of complex functions, the parametric function is plotted on the complex plane.

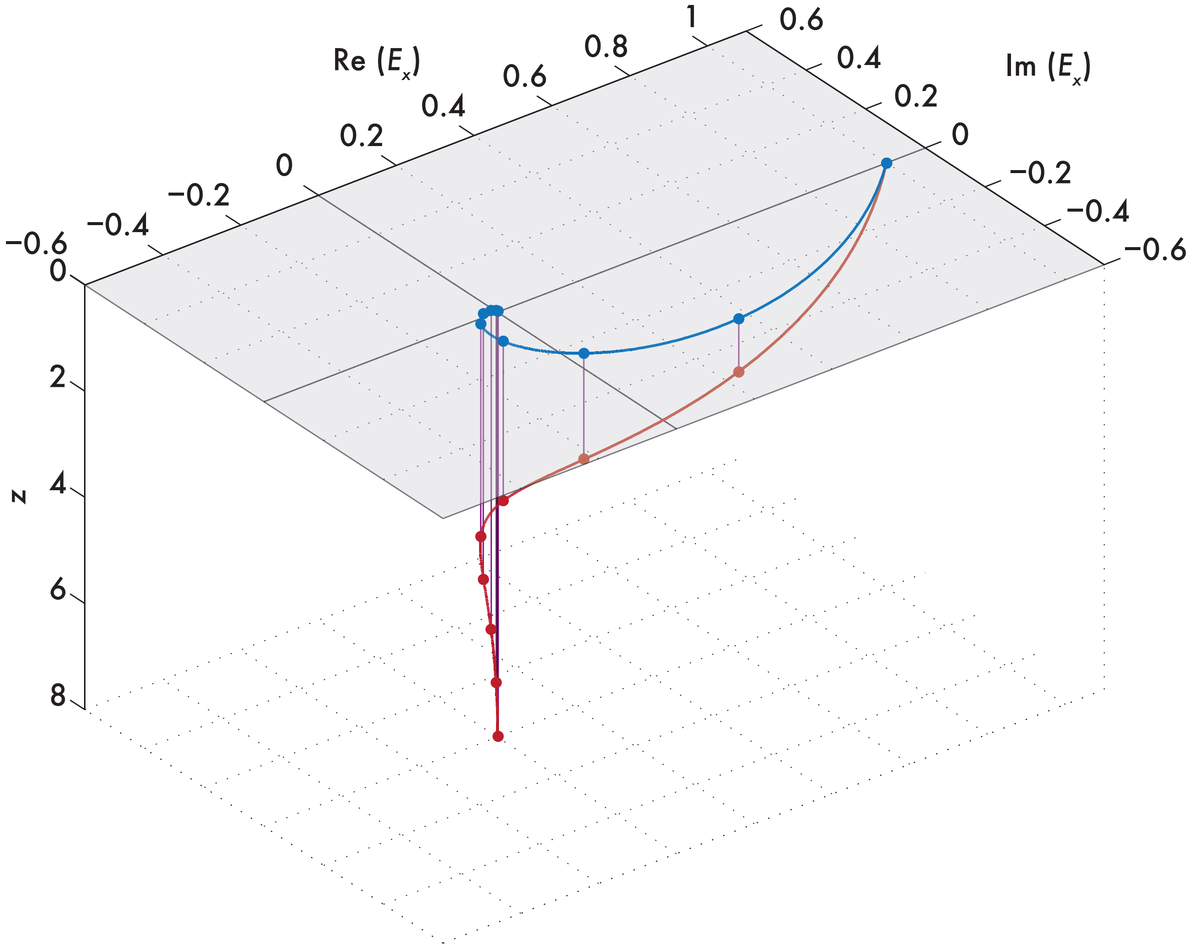

Figure 5: A complex function (Equation 12) of z. E_0 is assumed to be 1. The real and imaginary parts as a function of z are shown in red and as a projection onto the complex plane in blue. Figure 6 shows the projection onto the complex plane.

We've already discussed the complex plane. Plotting a complex function is no different than plotting an ordinary set of x and y components, in this case real and imaginary. The difference being

[1] Parametric equations are ones in which the coordinates are represented as functions of a single variable. Typically, these are represented in polar notation. For example, for a circle with radius r, $r^2 = x^2 + y^2$, can be represented parametrically as a function of a single parameter, θ,

$$x = r\cos\theta$$
$$y = r\sin\theta.$$

the representation of the function not as a typical input versus output but as a parametric function, that is

$$x = \mathrm{Re}(E_x(z))$$
$$y = \mathrm{Im}(E_x(z)),$$

where x are the real values of the function and y are the imaginary values of the function. One approach is to rewrite our equation in $x + iy$ form so that the real and imaginary parts can be easily separated, which we can do via the trigonometric representation.

To write the function in Equation 12 in $x + iy$ form,

$$\begin{aligned} E_x(z) &= E_0 \exp\left[-\frac{(1+i)}{\sqrt{2}} z\right] \\ &= E_0 \exp\left(-\frac{1}{\sqrt{2}} z\right) \exp\left(-\frac{i}{\sqrt{2}} z\right) \\ &= E_0 \exp\left(-\frac{1}{\sqrt{2}} z\right) \left[\cos\left(-\frac{1}{\sqrt{2}} z\right) + i \sin\left(-\frac{1}{\sqrt{2}} z\right)\right]. \end{aligned}$$

We can now define our parametric equations:

$$\mathrm{Re}(E_x(z)) = E_0 \exp\left(-\frac{1}{\sqrt{2}} z\right) \cos\left(-\frac{1}{\sqrt{2}} z\right), \tag{13}$$

$$\mathrm{Im}(E_x(z)) = E_0 \exp\left(-\frac{1}{\sqrt{2}} z\right) \sin\left(-\frac{1}{\sqrt{2}} z\right). \tag{14}$$

The parametric functions produce the projection of the spiral on the complex plane (Figure 6).

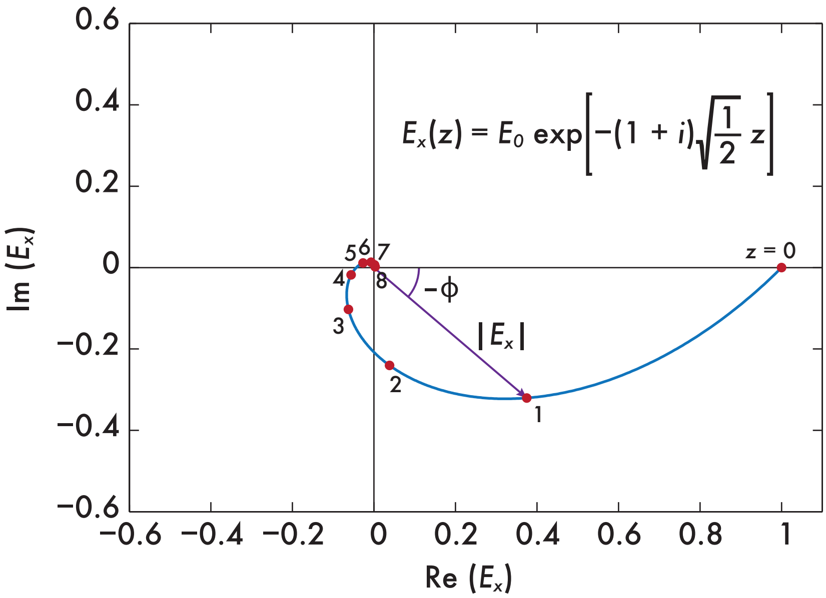

Figure 6: The parameterized plot of the complex function in Equation 12. The spiral nature of this equation is clearly evident in this plot. We can describe each point along the plot in terms of two parameters, a distance from the origin (norm) and an angle counterclockwise from the real axis (phase).

A complex function can sometimes be a bit difficult to interpret—we live in a real space, not an imaginary one—so it can be advantageous to write our equation as real numbers that we can ascribe physical meaning to our function, namely a magnitude and a phase. We have already found these parameters inadvertently on our way to a trigonometric representation,

$$|E_x(z)| = E_0 \exp\left[-\frac{z}{\sqrt{2}}\right], \tag{15}$$

$$\phi = -\frac{z}{\sqrt{2}}. \tag{16}$$

$$\tag{17}$$

The results are shown in Figure 7. Note 2π has been added to the phase so that it is positive. For

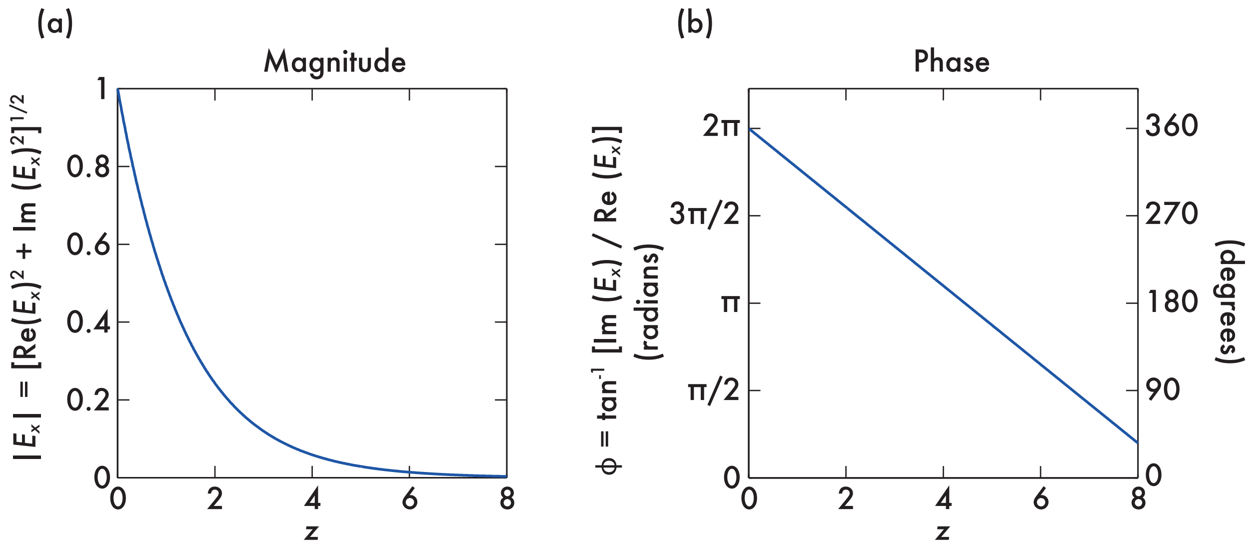

Figure 7: Representation of the complex function in Equation 12 as a magnitude (a) and phase (b) as a function of the independent variable z.

the parametric equations, the magnitude is given by

$$\begin{aligned}
|E_x(z)| &= (\mathrm{Re}(E_x)^2 + \mathrm{Im}(E_x)^2)^{1/2} \\
&= \left[E_0^2 \exp\left(-\frac{2}{\sqrt{2}}z\right)\cos^2\left(-\frac{1}{\sqrt{2}}z\right) + E_0^2 \exp\left(-\frac{2}{\sqrt{2}}z\right)\sin^2\left(-\frac{1}{\sqrt{2}}z\right)\right]^{1/2} \\
&= E_0 \exp\left(-\frac{1}{\sqrt{2}}z\right)\left[\cos^2\left(-\frac{1}{\sqrt{2}}z\right) + \sin^2\left(-\frac{1}{\sqrt{2}}z\right)\right]^{1/2} \\
&= E_0 \exp\left(-\frac{1}{\sqrt{2}}z\right)(1)^{1/2} \\
&= E_0 \exp\left(-\frac{1}{\sqrt{2}}z\right),
\end{aligned}$$

and the phase is given by

$$\begin{aligned}
\phi &= \tan^{-1}\left[\frac{\mathrm{Im}(E_x)}{\mathrm{Re}(E_x)}\right] \\
&= \tan^{-1}\left[\frac{E_0 \exp\left(-\frac{1}{\sqrt{2}}z\right)\sin\left(-\frac{1}{\sqrt{2}}z\right)}{E_0 \exp\left(-\frac{1}{\sqrt{2}}z\right)\cos\left(-\frac{1}{\sqrt{2}}z\right)}\right] \\
&= \tan^{-1}\left[\frac{\sin\left(-\frac{1}{\sqrt{2}}z\right)}{\cos\left(-\frac{1}{\sqrt{2}}z\right)}\right] \\
&= \tan^{-1}\left[\tan\left(-\frac{1}{\sqrt{2}}z\right)\right] \\
&= -\frac{1}{\sqrt{2}}z.
\end{aligned}$$

8 Derivatives of Complex Functions

I won't have you perform any derivatives of complex functions as part of the course. But for those that are interested, I've included this brief (non-comprehensive) section that will give you a basis in the idea of complex derivatives. Taking derivatives of complex functions is not necessarily much more complex than derivatives of real-valued functions. The sum, product, chain rule, etc. all still hold. However, there are extra conditions that must be met in order for the derivative to exist.

Consider a complex function, $f(z) = u(x, y) + iv(x, y)$, which we construct using two functions u and v. Recall from our calculus discussions that the derivative at a point, z_0, is defined as

$$\left.\frac{d}{dz}f(z)\right|_{z=z_0} = \lim_{\Delta z\to 0}\frac{f(z_0+\Delta z)-f(z_0)}{\Delta z}$$

Since $\Delta z = \Delta x + i\Delta y$, $\Delta z \to 0$ can occur by either $\Delta x \to 0$ or $i\Delta y \to 0$. Hence the derivative of a complex function, $f(z)$, may be performed by either

$$\frac{d}{dz}f(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}, \tag{18}$$

$$\frac{d}{dz}f(z) = -i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}. \tag{19}$$

$$\tag{20}$$

These equations leads to the Cauchy-Riemann equations,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \tag{21}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \tag{22}$$

implying that we have somewhat of a choice in how we determine the derivative of our complex function. The Cauchy-Riemann equations also show that the functions u and v must be related in this manner at every point z_0, and if not, the function is not differentiable.

Let's take a look at e^z and show that it's derivative is in fact itself. Using the standard from for a complex number,

$$\begin{aligned}
e^z &= e^{x+iy} \\
&= e^x e^{iy} \\
&= e^x(\cos y + i \sin y) \\
&= e^x \cos y + e^x i \sin y.
\end{aligned}$$

Since we can choose x or y to take our derivative, we'll take x for simplicity first,

$$\begin{aligned}
\frac{d}{dz}e^z &= \frac{\partial}{\partial x}(e^x \cos y) + i\frac{\partial}{\partial x}(e^x \sin y) \\
&= e^x \cos y + ie^x \sin y \\
&= e^z.
\end{aligned}$$

But we aren't done. We must show that the Cauchy-Riemann equations hold or that both relationships in Equations 18,

$$\begin{aligned}
\frac{\partial}{\partial x}(e^x \cos y) &= \frac{\partial}{\partial y}[e^x \sin y] \\
e^x \cos y &= e^x \cos y,
\end{aligned}$$

and Equation 20 yield the same result,

$$\begin{aligned}
\frac{\partial}{\partial y}(e^x \cos y) &= -\frac{\partial}{\partial x}[e^x \sin y] \\
-e^x \sin y &= -e^x \sin y.
\end{aligned}$$

Since the Cauchy-Riemann conditions are met and e^z is continuous, the derivative of e^z is in fact itself.